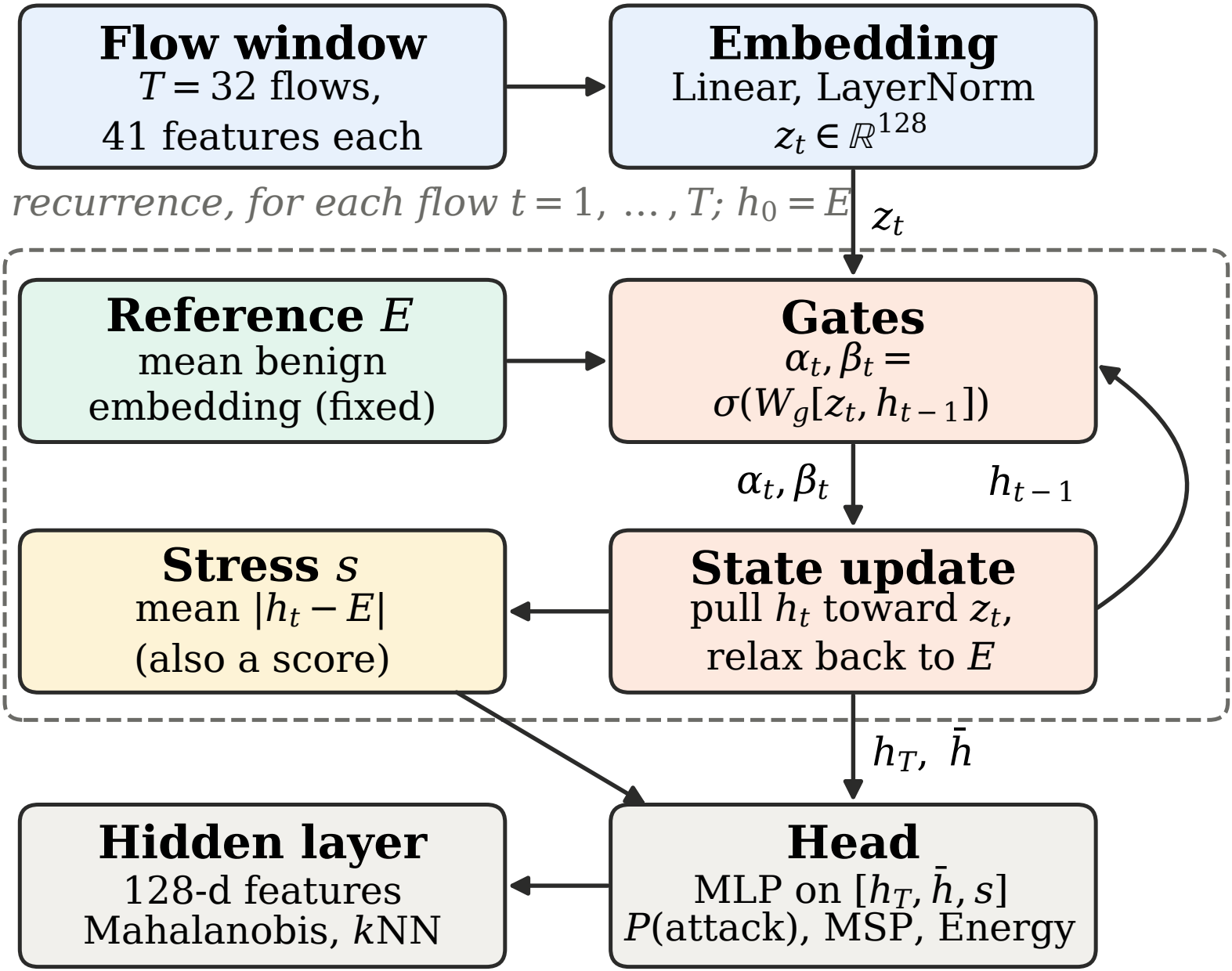
Flow window
$T = 32$ flows,
41 features each
Embedding
Linear, LayerNorm
$z_t \in \mathbb{R}^{128}$
recurrence, for each flow $t = 1, \ldots, T$; $h_0 = E$
z_t
Reference E
mean benign
embedding (fixed)
Gates
$\alpha_t, \beta_t =$
$\sigma(W_g[z_t, h_{t-1}])$
α_t, β_t
h_{t-1}
Stress s
mean $|h_t - E|$
(also a score)
State update
pull h_t toward z_t,
relax back to E
$h_T,\ \bar{h}$
Hidden layer
128-d features
Mahalanobis, kNN
Head
MLP on $[h_T, \bar{h}, s]$
P(attack), MSP, Energy